EE 66 Signals, Dynamics, and Information
Fall 2026 Note 8

8.1 Singular Value Decomposition

For the past few weeks we've been talking about solving linear systems of equations. For instance, we might have a system $\mathbf{Ax} = \mathbf{b}$ where we want to find $\mathbf{b}$. Using that same equation, we can look at it another way: sometimes we want to solve for $\mathbf{x}$ given an output vector $\mathbf{b}$ and a matrix of data points $\mathbf{A}$. That should be familiar to you from the context of matrix inverses and QR decomposition manipulations. But we've mostly been looking at square matrices when solving such problems. What if we wanted to solve $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{A}$ is non-square? If you recall, only square matrices have true inverses, so how would that even work?

Enter, the Singular Value Decomposition, or SVD. Just like how QR "factors" a matrix into components that we can easily manipulate, SVD does the same thing, but more: the SVD can also give insight into the structure of our data. We use the SVD to invert non-square matrices, pare down large datasets, and pick out important features in machine learning. For those of you interested in Data Science, SVD is one of the most effective algorithms for building recommendation systems (think Netflix movies, YouTube videos, and Spotify songs), and if you know a bit of Python, you can build your own recommendation systems!

8.1.1 Spectral Theorem

The SVD relies heavily on the Spectral Theorem from Note 7, which asserts the following:

Theorem 8.1: Spectral Theorem: Let $\mathbf{S} \in \mathbb{R}^{n \times n}$ be symmetric, i.e., $\mathbf{S} = \mathbf{S}^\top$. Then

(i) All eigenvalues of $\mathbf{S}$ are real.

(ii) $\mathbf{S}$ has n orthonormal eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$. Equivalently, $\mathbf{S} = \mathbf{V}\Lambda\mathbf{V}^\top$, where $\mathbf{V} = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix}$ is orthonormal and Λ is diagonal with the eigenvalues of $\mathbf{S}$ on its diagonal.

In particular, every symmetric matrix is diagonalizable.

8.1.2 Full SVD

Definition 8.1 (Full SVD): Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have rank $r \leq \min\{m, n\}$. A *(full) SVD* of $\mathbf{A}$ is a decomposition

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top \tag{1}$$

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where

(i) $\mathbf{U} = \begin{bmatrix}\mathbf{U}_r & \mathbf{U}_{m-r}\end{bmatrix} \in \mathbb{R}^{m\times m}$ is a square orthonormal matrix whose columns are the so-called *left singular vectors* of $\mathbf{A}$; here $\mathbf{U}_r \in \mathbb{R}^{m\times r}$ and $\mathbf{U}_{m-r} \in \mathbb{R}^{m\times(m-r)}$ are tall matrices with orthonormal columns.

(ii) $\mathbf{V} = \begin{bmatrix}\mathbf{V}_r & \mathbf{V}_{n-r}\end{bmatrix} \in \mathbb{R}^{n\times n}$ is a square orthonormal matrix whose columns are the so-called *right singular vectors* of $\mathbf{A}$; here $\mathbf{V}_r \in \mathbb{R}^{n\times r}$ and $\mathbf{V}_{n-r} \in \mathbb{R}^{n\times(n-r)}$ are tall matrices with orthonormal columns.

(iii) $\Sigma = \begin{bmatrix}\Sigma_r & 0_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)}\end{bmatrix} \in \mathbb{R}^{m\times n}$ is a non-square diagonal matrix whose diagonal entries $\sigma_1 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = \sigma_{\min\{m,n\}} = 0$ are non-negative ordered real numbers, and are the so-called *singular values* of A.

First, notice that m and n are not necessarily equal, meaning SVD works for any matrix, not just for square ones. Second, notice that $\mathbf{U}$ and $\mathbf{V}$ are orthonormal matrices, which we've seen are very nice to work with. We now present some useful linear algebra properties from the SVD.

Theorem 8.2: Linear Algebra of the Full SVD: Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have rank $r \leq \min\{m,n\}$. Let $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$ be an SVD of $\mathbf{A}$.

(i) $\text{Col}(\mathbf{U_r}) = \text{Col}(\mathbf{A})$.

(ii) $\text{Col}(\mathbf{V_r}) = \text{Col}(\mathbf{A}^\top)$.

(iii) $\text{Col}(\mathbf{U_{m-r}}) = \text{Null}(\mathbf{A}^\top)$.

(iv) $\text{Col}(\mathbf{V_{n-r}}) = \text{Null}(\mathbf{A})$.

(v) $\mathbf{A}\mathbf{A}^\top = \mathbf{U}\Sigma\Sigma^\top\mathbf{U}^\top$.

(vi) $\mathbf{A}^\top\mathbf{A} = \mathbf{V}\Sigma^\top\Sigma\mathbf{V}^\top$.

(vii) $\mathbf{A}\mathbf{V}_r = \mathbf{U}_r\Sigma_r$.

The proof of the Linear Algebra of the Full SVD is on the longer side and may distract from the overall flow of this note, so it is left to the appendix.

What do these properties mean? The first property takes the first r columns of the $\mathbf{U}$ matrix and claims they span the same vector space as the columns of $\mathbf{A}$. What that means is we can boil down the column space of $\mathbf{A}$ to $\mathbf{U}_r$, where $\mathbf{U}_r$ is much easier to work with since the columns of this matrix are orthonormal. You can draw a similar conclusion about property (ii) with respect to the row space of $\mathbf{A}$.

Where things get confusing is properties (iii) and (iv). Why are the columns of $\mathbf{U}_r$ related to the columns of $\mathbf{V}_{n-r}$, even though $\mathbf{V}_{n-r}$ is related to the row space of $\mathbf{A}$? By the rank nullity theorem, $\dim(\text{Col}(\mathbf{A})) + \dim(\text{Null}(\mathbf{A})) = n$, the

number of columns. Since $\text{Col}(\mathbf{A}) \equiv \text{Col}(\mathbf{U}_r)$, and $\text{Null}(\mathbf{A}) \equiv \text{Col}(\mathbf{V}_{n-r})$, the dimensions work out. Mathematically, we show that:

$$\begin{aligned} \mathbf{A} &= \mathbf{U}\Sigma\mathbf{V}^\top \\ \mathbf{AV} &= \mathbf{U}\Sigma\mathbf{V}^\top\mathbf{V} \\ \mathbf{AV} &= \mathbf{U}\Sigma \end{aligned} \tag{2}$$

Next, we consider $\mathbf{V}$ as a block matrix of $\mathbf{V}_r$ and $\mathbf{V}_{n-r}$. Looking only at the last $n-r$ columns of $\mathbf{V}$, that means we only look at the last $n-r$ singular values of Σ, which are 0.

$$\mathbf{A}\mathbf{v}_i = 0 \quad \forall\, i \in [r+1,\, n] \tag{3}$$

We've shown that the last $n-r$ columns of $\mathbf{V}_{n-r}$ are in the null space of $\mathbf{A}$.

But conceptually, this is quite difficult to grapple with, and is discussed in more detail in the Full SVD Intuition section.

Properties (v) and (vi) become relevant when talking about inverting non-square matrices, which will be covered in the next note!

Now that we know how useful the SVD is, how can we factorize an arbitrary matrix $\mathbf{A}$ into its SVD form?

8.1.3 Finding the SVD

Let's walk through the process of finding the SVD for an arbitrary matrix $A \in \mathbb{R}^{m\times n}$:

1. Compute the symmetric matrix: First, find either the matrix $A^\top A$ (Case I) or $AA^\top$ (Case II). Choose whichever product gives you a smaller matrix.
2. Compute the eigenvector-eigenvalue pairs of your chosen symmetric matrix.
3. Construct V or U:
    - (Case I) Normalize each eigenvector to form the columns of V.
    - (Case II) Normalize each eigenvector to form the columns of U.
    - For both cases, if you don't have enough eigenvectors to span the entire space (n for V, m for U), extend basis. Add in unit vectors and apply Gram Schmidt to ensure orthonormality.
4. Construct Σ: Take the square root of each eigenvalue, and place them along the diagonal of your Σ matrix in descending order. Fill the remaining entries with zeros.
5. Construct U or V:

- (Case I) Each column vector of U is given by $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$
- (Case II) Each column vector of V is given by $\mathbf{v}_i = \frac{1}{\sigma_i} A^\top \mathbf{u}_i$

6. Write out the full SVD: Take the transpose of your V matrix, then write $A = U\Sigma V^\top$. Make sure to order the columns of your eigenvectors in U and V to match with their eigenvalues. For V, since you eventually transpose it, notice the first singular value maps to the first row of $V^\top$, and so on.

Theorem 8.3: Correctness of Construction of Full SVD: Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have rank $r \leq \min\{m,n\}$. Let $\mathbf{U},\Sigma,\mathbf{V} :=$ FULLSVD$(\mathbf{A})$ be the output of the full SVD construction algorithm. Then $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$ is a full SVD of $\mathbf{A}$.

The proof of the Correctness of Construction of Outer Product Form of SVD is on the longer side and may distract from the overall flow of this note, but we strongly encourage you to read through the proof. We've provided two alternate methods of showing the same result. Both build the same SVD and then verify it, but the first argues one singular vector at a time, while the second reasons about entire matrices at once and derives the formula for $\mathbf{U}$ *instead of taking it as given.*

Why do we use $A^\top A$ and $AA^\top$ for the SVD? It turns out that they have special properties with respect to the eigenvalues and eigenvectors!

Let $A \in \mathbb{R}^{m\times n}$ have rank $r \leq \min\{m,n\}$. Then $A^\top A \in \mathbb{R}^{n\times n}$ and $AA^\top \in \mathbb{R}^{m\times m}$ are symmetric matrices of rank r. Each has exactly r nonzero eigenvalues, which are real and positive.

See the appendix for proof.

$A^\top A$ and $AA^\top$ generate eigenvalues that we can turn into our Σ. The corresponding eigenvectors are orthogonal, which we leveraged to generate U and $V^\top$!

8.1.4 Compact SVD

What if we didn't care about null spaces? We can also rewrite the SVD in a compact form, representing only the linearly independent rows or columns of our matrix $\mathbf{A}$.

Definition 8.2 (Compact SVD): Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have rank $r \leq \min\{m,n\}$. A *compact SVD* of $\mathbf{A}$ is a decomposition

$$\mathbf{A} = \mathbf{U}_r \Sigma_r \mathbf{V}_r^\top \tag{4}$$

where

(i) $\mathbf{U}_r \in \mathbb{R}^{m\times r}$ is a matrix with orthonormal columns, which are so-called *left singular vectors* of $\mathbf{A}$.

(ii) $\mathbf{V}_r \in \mathbb{R}^{n\times r}$ is a matrix with orthonormal columns, which are so-called *right singular vectors* of $\mathbf{A}$.

(iii) $\Sigma_r \in \mathbb{R}^{r\times r}$ is a diagonal matrix whose diagonal entries $\sigma_1 \geq \cdots \geq \sigma_r > 0$ are positive ordered real numbers, and are so-called *singular values* of A.

8.1.5 Finding the Compact SVD

To find the Compact SVD, follow the same process as the full SVD, except all matrices have r columns, where r is the rank of A.

The compact SVD is connected to the full SVD via the following calculation.

$$\begin{aligned}
\mathbf{U}\Sigma\mathbf{V}^\top &= \begin{bmatrix}\mathbf{U}_r & \mathbf{U}_{m-r}\end{bmatrix}\begin{bmatrix}\Sigma_r & \mathbf{0}_{r\times(n-r)} \\ \mathbf{0}_{(m-r)\times r} & \mathbf{0}_{(m-r)\times(n-r)}\end{bmatrix}\begin{bmatrix}\mathbf{V}_r & \mathbf{V}_{n-r}\end{bmatrix}^\top \\
&= \begin{bmatrix}\mathbf{U}_r & \mathbf{U}_{m-r}\end{bmatrix}\begin{bmatrix}\Sigma_r & \mathbf{0}_{r\times(n-r)} \\ \mathbf{0}_{(m-r)\times r} & \mathbf{0}_{(m-r)\times(n-r)}\end{bmatrix}\begin{bmatrix}\mathbf{V}_r^\top \\ \mathbf{V}_{n-r}^\top\end{bmatrix} \\
&= \begin{bmatrix}\mathbf{U}_r & \mathbf{U}_{m-r}\end{bmatrix}\left(\begin{bmatrix}\Sigma_r \\ \mathbf{0}_{(m-r)\times r}\end{bmatrix}\mathbf{V}_r^\top + \begin{bmatrix}\mathbf{0}_{r\times(n-r)} \\ \mathbf{0}_{(m-r)\times(n-r)}\end{bmatrix}\mathbf{V}_{n-1}^\top\right) \\
&= \begin{bmatrix}\mathbf{U}_r & \mathbf{U}_{m-r}\end{bmatrix}\begin{bmatrix}\Sigma_r \\ \mathbf{0}_{(m-r)\times r}\end{bmatrix}\mathbf{V}_r^\top \\
&= \begin{bmatrix}\mathbf{U}_r & \mathbf{U}_{m-r}\end{bmatrix}\begin{bmatrix}\Sigma_r\mathbf{V}_r^\top \\ \mathbf{0}_{(m-r)\times n}\end{bmatrix} \\
&= \mathbf{U}_r\Sigma_r\mathbf{V}_r^\top + \mathbf{U}_{m-r}\mathbf{0}_{(m-r)\times n} \\
&= \mathbf{U}_r\Sigma_r\mathbf{V}_r^\top. \qquad (5)
\end{aligned}$$

Thus, the sub-matrices of a full SVD are a compact SVD. The compact SVD has the same properties as the full SVD, except without the null spaces.

Theorem 8.4: Linear Algebra of the Compact SVD: Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have rank $r \leq \min\{m,n\}$. Let $\mathbf{A} = \mathbf{U}_r\Sigma_r\mathbf{V}_r^\top$ be an compact form SVD of $\mathbf{A}$.

(i) $\text{Col}(\mathbf{U_r}) = \text{Col}(\mathbf{A})$.

(ii) $\text{Col}(\mathbf{V_r}) = \text{Col}(\mathbf{A}^\top)$.

(iii) $\mathbf{A}\mathbf{A}^\top\mathbf{U}_r = \mathbf{U}_r\Sigma_r^2$.

(iv) $\mathbf{A}^\top\mathbf{A}\mathbf{V}_r = \mathbf{V}_r\Sigma_r^2$.

(v) $\mathbf{A}\mathbf{V}_r = \mathbf{U}_r\Sigma_r$.

Check your understanding: Prove the takeaways from the Linear Algebra of the Compact SVD theorem.

Theorem 8.5: Correctness of Construction of the Compact SVD: Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have rank $r \leq \min\{m,n\}$. Let $\mathbf{U}_r, \Sigma_r, \mathbf{V}_r := \text{COMPACTSVD}(\mathbf{A})$ be the output of the compact SVD construction algorithm. Then $\mathbf{A} = \mathbf{U}_r\Sigma_r\mathbf{V}_r^\top$ is a compact SVD of $\mathbf{A}$.

Check your understanding: Prove the Correctness of the Construction of the Compact SVD.

8.1.6 Outer Product Form of the SVD

Now what if we wanted to break down the SVD in terms of individual vectors? Outer Product Form does just that!

Definition 8.3 (Outer Product Form of the SVD): Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have rank $r \leq \min\{m,n\}$. An *outer product form of an SVD* of $\mathbf{A}$ is a decomposition

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \tag{6}$$

where

(i) $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\} \subseteq \mathbb{R}^m$ is an orthonormal set of vectors, and are so-called *left singular vectors* of $\mathbf{A}$.

(ii) $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\} \subseteq \mathbb{R}^n$ is an orthonormal set of vectors, and are so-called *right singular vectors* of $\mathbf{A}$.

(iii) $\sigma_1 \geq \cdots \geq \sigma_r > 0$ are positive ordered scalars, and are so-called *singular values* of $\mathbf{A}$.

While this form looks quite unassuming, it actually reveals several linear-algebraic properties of $\mathbf{A}$, as we will see in the following theorem.

Theorem 8.6: Linear Algebra of the Outer Product SVD Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have rank $r \leq \min\{m,n\}$. Let $\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ be an outer product form SVD of $\mathbf{A}$.

(i) $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ is an orthonormal basis for $\text{Col}(\mathbf{A})$.

(ii) $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is an orthonormal basis for $\text{Col}(\mathbf{A}^\top)$.

(iii) For each i, $1 \leq i \leq r$, we have $(\sigma_i^2, \mathbf{u}_i)$ is an eigenvalue-eigenvector pair for $\mathbf{A}\mathbf{A}^\top$.

(iv) For each i, $1 \leq i \leq r$, we have $(\sigma_i^2, \mathbf{v}_i)$ is an eigenvalue-eigenvector pair for $\mathbf{A}^\top\mathbf{A}$.

(v) For each i, $1 \leq i \leq r$, we have $\mathbf{A}\mathbf{v}_i = \sigma_i \mathbf{u}_i$.

The proof of the Linear Algebra of the Outer Product SVD is on the longer side and may distract from the overall flow of this note, so it is left to the appendix.

8.1.7 Finding the Outer Product Form of SVD

To find the Outer Product form of SVD, follow the same process as the full SVD, except represent all matrices as a summation of vector multiplications. Outer Product form is easier to write and compute by hand.

Theorem 8.7: Correctness of Outer Product Form Algorithm : Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have rank $r \leq \min\{m,n\}$. Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}, \{\sigma_1, \ldots, \sigma_r\}, \{\mathbf{v}_1, \ldots, \mathbf{v}_r\} := \text{OUTERPRODUCTSVD}(A)$ be the output of the outer product SVD construction algorithm. Then $\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ is an outer product SVD of $\mathbf{A}$.

The proof of the Correctness of the Outer Product Form Algorithm is on the longer side and may distract from the overall flow of this note, so it is left to the appendix.

The Outer Product form of SVD is connected to the Compact SVD by the following calculation.

$$\begin{aligned}\mathbf{U}_r\Sigma_r\mathbf{V}_r^\top = [\mathbf{u}_1\cdots\mathbf{u}_r]\begin{bmatrix}\sigma_1 & & \\ & \ddots & \\ & & \sigma_r\end{bmatrix}[\mathbf{v}_1\cdots\mathbf{v}_r]^\top &= [\mathbf{u}_1\cdots\mathbf{u}_r]\begin{bmatrix}\sigma_1 & & \\ & \ddots & \\ & & \sigma_r\end{bmatrix}\begin{bmatrix}\mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_r^\top\end{bmatrix} \\ &= [\mathbf{u}_1\cdots\mathbf{u}_r]\begin{bmatrix}\sigma_1\mathbf{v}_1^\top \\ \vdots \\ \sigma_r\mathbf{v}_r^\top\end{bmatrix} \\ &= \sum_{i=1}^{r}\sigma_i\mathbf{u}_i\mathbf{v}_i^\top. \end{aligned} \tag{7}$$

8.1.8 Comparison Between the SVD Forms

We identify weaknesses and strengths of using the SVD forms, so you can decide which one to use.

- Full SVD:
  - Weaknesses: It is the most computationally intensive to compute, either by computer or by hand. It requires running Gram-Schmidt twice to find $\mathbf{U}_{m-r}$ and $\mathbf{V}_{n-r}$. Also Σ is non-square and thus not invertible.
  - Strengths: The matrices $\mathbf{U}$ and $\mathbf{V}$ are square orthonormal, and thus invertible. There is also a characterization of null spaces of $\mathbf{A}^\top$ and $\mathbf{A}$ as the column spaces of $\mathbf{U}_{m-r}$ and $\mathbf{V}_{n-r}$ respectively.
- Compact SVD:
  - Weaknesses: The matrices $\mathbf{U}_r$ and $\mathbf{V}_r$ are not square, so they do not have an inverse. Also, there is no characterization of null spaces, as in the full SVD.
  - Strengths: The matrix Σ_r is square and invertible. It is also easier to construct by hand than the full SVD.
- Outer Product SVD:
  - Weaknesses: The summation notation is messy and sometimes tedious to work with. Also, there is no characterization of null spaces, as in the full SVD.
  - Strengths: It is the most computationally efficient to construct by computer, and also saves the most memory. It is also easier to construct by hand than the full SVD.

When using the SVD to compute things, one potential rule to use is to use the compact SVD unless there is a need to analyze null spaces, in which case the full SVD is essentially required.

8.1.9 Full SVD Intuition

In the above description, we mentioned that $\text{Null}(\mathbf{A}) = \text{Col}(\mathbf{V_{n-r}})$ and $\text{Null}(\mathbf{A}^\top) = \text{Col}(\mathbf{U_{m-r}})$. Why does this make sense?

First of all, recall that null spaces are subsets of *input* spaces. Thus, since $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$, the input space of $\mathbf{A}$ must match the dimension of the column vectors of $\mathbf{V}$, so it makes sense that $\text{Null}(\mathbf{A})$ is related to the column vectors of $\mathbf{V}$. Similarly $\mathbf{A}^\top = \mathbf{V}\Sigma^\top\mathbf{U}^\top$ so $\text{Null}(\mathbf{A}^\top)$ makes sense to be related to the column vectors of $\mathbf{U}$.

To explain the specific submatrices of $\mathbf{V}$ and $\mathbf{U}$ that describe the null spaces, let's examine the outer product form of SVD (for now, let's just focus on the matrix $\mathbf{A}$):

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$$

Now, let's see what happens if we multiply one of the "extra" $\mathbf{v}_j$ vectors (which correspond to $j = r+1, \ldots, n$, the column vectors of $\mathbf{V}_{n-r}$) by the matrix $\mathbf{A}$:

$$\mathbf{A}\mathbf{v}_j = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \mathbf{v}_j$$

Since $i \neq j$ for all terms in this summation, $\mathbf{v}_i^\top \mathbf{v}_j = 0$ (since $\{\mathbf{v}_i\}$ is an orthonormal set, by construction).

Thus:

$$\mathbf{A}\mathbf{v}_j = 0$$

for $j = r+1, \ldots, n$, which means that these $\mathbf{v}_j$ are in $\text{Null}(\mathbf{A})$.

By linearity, every linear combination of $\mathbf{v}_j$ is also in $\text{Null}(\mathbf{A})$. By rank-nullity theorem, since $\text{rank}(\mathbf{A}) = r$ and A has n columns, $\dim(\text{Null}(\mathbf{A})) = n - r$, which is exactly the number of column vectors in $\mathbf{V}_{n-r}$! Since we already established that the column vectors of $\mathbf{V}_{n-r}$ are in $\text{Null}(\mathbf{A})$, this implies that:

$$\text{Null}(\mathbf{A}) = \text{Col}(\mathbf{V_{n-r}}) \tag{8}$$

Symmetrically, the outer product SVD of $\mathbf{A}^\top$ is:

$$\mathbf{A}^\top = \sum_{i=1}^{r} \sigma_i \mathbf{v}_i \mathbf{u}_i^\top$$

With the same logic flow as $\mathbf{A}$:

$$\text{Null}(\mathbf{A}^\top) = \text{Col}(\mathbf{U_{m-r}}) \tag{9}$$

8.2 Singular Value Decomposition: Geometric Properties

Let $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$ be the SVD of some matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$.

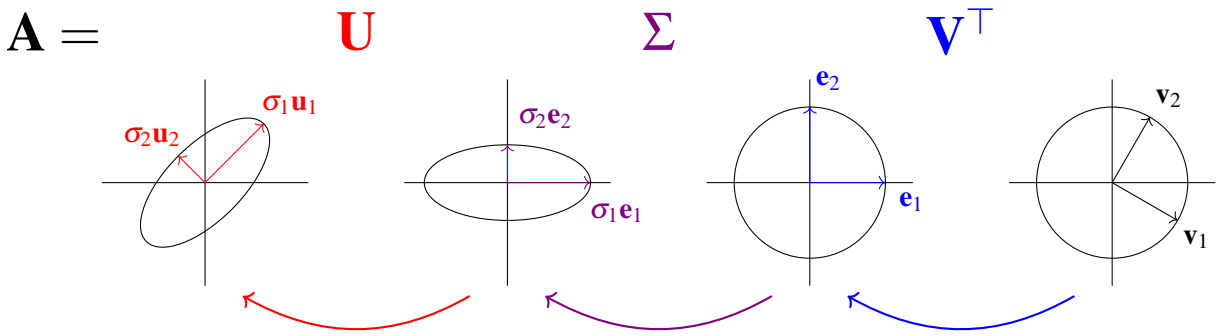
Recall that for any vector $\mathbf{x}$ and any orthogonal matrix $\mathbf{U}$ that $\|\mathbf{Ux}\| = \|\mathbf{x}\|$. Thus we can interpret multiplication by an orthonormal matrix as a combination of operations that don't change length, such as rotations and reflections.

Since Σ_r is diagonal with entries $\sigma_1, \ldots, \sigma_r$, multiplying a vector by Σ stretches the first entry of the vector by σ_1, the second entry by σ_2, and so on.

Combining these observations, we interpret $\mathbf{Ax}$ as the composition of three operations:

1. $\mathbf{V}^\top\mathbf{x}$ which rotates $\mathbf{x}$ without changing its length.
2. $\Sigma\mathbf{V}^\top\mathbf{x}$ which stretches the resulting vector along each axis with the corresponding singular value, and may also collapse or add new dimensions,
3. $\mathbf{U}\Sigma\mathbf{V}^\top\mathbf{x}$ which again rotates the resulting vector without changing its length.

The following figure illustrates these three operations moving from the right to the left.



Here as usual $\mathbf{e}_1, \mathbf{e}_2$ are the first and second standard basis vectors.

The geometric interpretation above reveals that σ_1 is the largest amplification factor a vector can experience upon multiplication by $\mathbf{A}$. More specifically, if $\|\mathbf{x}\| \leq 1$ then $\|\mathbf{Ax}\| \leq \sigma_1$. We achieve equality at $\mathbf{x} = \mathbf{v}_1$, because then

$$\|\mathbf{Ax}\| = \left\|\mathbf{U}\Sigma\mathbf{V}^\top\mathbf{v}_1\right\| = \|\mathbf{U}\Sigma\mathbf{e}_1\| = \|\sigma_1\mathbf{U}\mathbf{e}_1\| = \|\sigma_1\mathbf{u}_1\| = \sigma_1\|\mathbf{u}_1\| = \sigma_1. \tag{10}$$

8.3 Examples

8.3.1 Example SVD Interpretation and Application

Suppose we want to build a least squares regression model to predict whether users like a movie they've never seen before, based on attributes of movies they have seen. We have an $m \times n$ matrix $\mathbf{A}$ of rank r that contains the ratings

of m viewers for n movies. Write

$$A = U\Sigma V^\top = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top.$$

We can interpret each rank 1 matrix $\sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ to be due to a particular attribute, e.g., comedy, action, sci-fi, or romance content. Then σ_i determines how strongly the ratings depend on the ith attribute; the entries of $\mathbf{v}_i^\top$ score each movie with respect to this attribute, and the entries of $\mathbf{u}_i$ evaluate how much each viewer cares about this particular attribute.

Interestingly, the $(r+1)^{\text{th}}$ attributes onwards don't influence the ratings, according to our analysis. This means the SVD can reveal which attributes actually matter in our model, allowing us to build simpler models (saving compute), and also ensures we actually can use our model. To see why, suppose we have an attribute with a singular value of 0. The corresponding column and row in $\mathbf{A}$ doesn't add any new information (since they lie beyond the rank of our matrix), and actually makes our data matrix collapse such that we can't perform least squares regression.

Now, let's dive into some real numbers!

8.3.2 Numerical Example 1

Let's find the SVD for

$$\mathbf{A} = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}.$$

We find the SVD for $\mathbf{A}^\top$ first and then take the transpose. We calculate

$$(\mathbf{A}^\top)^\top(\mathbf{A}^\top) = \mathbf{A}\mathbf{A}^\top = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}.$$

This happens to be diagonal[1], so we can read off the eigenvalues:

$$\lambda_1 = 32 \qquad \lambda_2 = 18$$

We can select the orthonormal eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The singular values are

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{32} = 4\sqrt{2} \qquad \sigma_2 = \sqrt{\lambda_2} = \sqrt{18} = 3\sqrt{2}.$$

Then to find $\mathbf{v}_1, \mathbf{v}_2$, we do

$$\mathbf{v}_1 = \frac{A^\top \mathbf{u}_1}{\sigma_1} = \frac{1}{4\sqrt{2}}\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$
$$\mathbf{v}_2 = \frac{A^\top \mathbf{u}_2}{\sigma_2} = \frac{1}{3\sqrt{2}}\begin{bmatrix} -3 \\ 3 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}.$$

[1] in cases where $\mathbf{A}^\top\mathbf{A}$ or $\mathbf{A}\mathbf{A}^\top$ aren't so nice, we'd solve for the eigenvalues and eigenvectors using $(\lambda\mathbf{I} - \mathbf{A})\mathbf{v} = 0$

Thus our SVD is

$$A = U\Sigma V^\top = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4\sqrt{2} & 0 \\ 0 & 3\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}. \tag{11}$$

Note that we can change the signs of $\mathbf{u}_1, \mathbf{u}_2$ and they are still orthonormal eigenvectors, and produce a valid SVD. However, changing the sign of $\mathbf{u}_i$ requires us to change the sign of $\mathbf{v}_i = \mathbf{A}^\top \mathbf{u}_i$, so therefore the product of $\mathbf{u}_i \mathbf{v}_i^\top$ remains unchanged.

Another source of non-uniqueness arises when we have repeated singular values, as seen in the next example.

8.3.3 Numerical Example 2

We want to find an SVD for

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Again, we find the SVD for $\mathbf{A}^\top$ and then take the transpose. Note that $\mathbf{A}\mathbf{A}^\top = \mathbf{I}_2$, which has repeated eigenvalues at $\lambda_1 = \lambda_2 = 1$. In particular, *any* pair of orthonormal vectors is a set of orthonormal eigenvectors for $\mathbf{I}_2 = \mathbf{A}\mathbf{A}^\top$. We can parameterize all such pairs as

$$\mathbf{u}_1 = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix} \qquad \mathbf{u}_2 = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}$$

where θ is a free parameter. Since $\sigma_1 = \sigma_2 = 1$, we obtain

$$\mathbf{v}_1 = \frac{\mathbf{A}^\top \mathbf{u}_1}{\sigma_1} = \begin{bmatrix} \cos(\theta) \\ -\sin(\theta) \end{bmatrix} \qquad \mathbf{v}_2 = \frac{\mathbf{A}^\top \mathbf{u}_2}{\sigma_2} = \begin{bmatrix} -\sin(\theta) \\ -\cos(\theta) \end{bmatrix}.$$

Thus an SVD is

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ -\sin(\theta) & -\cos(\theta) \end{bmatrix} \tag{12}$$

for any value of θ. Thus this matrix has *infinite* valid SVDs, one for each value of θ in the interval $[0, 2\pi)$.

8.4 Final Comments

We introduced the singular value decomposition, walked through its three forms and geometric interpretation, and demonstrated a few hands on examples.

The SVD is a very powerful method to do efficient and expressive data analysis. It can work with any matrix and reveal the most important features in a data matrix. We will see more applications of this flavor in the next note.

A Proofs for SVD

A.1 Proof of Eigenvalues of $A^\top A$ and $AA^\top$

Proof. We have

$$(\mathbf{A}^\top \mathbf{A})^\top = (\mathbf{A})^\top (\mathbf{A}^\top)^\top = \mathbf{A}^\top A$$

so $(\mathbf{A}^\top \mathbf{A})^\top = \mathbf{A}^\top \mathbf{A}$, and thus $\mathbf{A}^\top \mathbf{A}$ is symmetric.

Next, we show that $\text{rank}(A^\top A) = \text{rank}(A) = r$ by showing that $\text{Null}(\mathbf{A}^\top \mathbf{A}) = \text{Null}(\mathbf{A})$. If $A\mathbf{v} = \mathbf{0}$, then $A^\top A\mathbf{v} = A^\top \mathbf{0} = \mathbf{0}$. Conversely, if $A^\top A\mathbf{v} = \mathbf{0}$, then pre-multiplying by $\mathbf{v}^\top$ gives

$$0 = \mathbf{v}^\top A^\top A\mathbf{v} = (A\mathbf{v})^\top (A\mathbf{v}) = \left\|A\mathbf{v}\right\|^2,$$

so $A\mathbf{v} = \mathbf{0}$. Since A and $A^\top A$ both have n columns and the same null space, the Rank-Nullity Theorem gives

$$\text{rank}\left(A^\top A\right) = n - \dim\left(\text{Null}(\mathbf{A}^\top \mathbf{A})\right) = n - \dim\left(\text{Null}(\mathbf{A})\right) = \text{rank}(A) = r. \tag{13}$$

Now we show that $A^\top A$ has exactly r nonzero eigenvalues. Since $A^\top A$ is symmetric, the Spectral Theorem says it is diagonalizable as $A^\top A = V\Lambda V^{-1}$. Multiplying by an invertible matrix does not change rank, so $r = \text{rank}\left(A^\top A\right) = \text{rank}(\Lambda)$, which is the number of nonzero diagonal entries of Λ, i.e., the number of nonzero eigenvalues of $A^\top A$ (counted with multiplicity).

We now show that all nonzero eigenvalues of $A^\top A$ are real and positive. Since $A^\top A$ is symmetric, the Spectral Theorem says that $A^\top A$ has real eigenvalues. It remains to show that they are all non-negative. Let λ be a nonzero eigenvalue. Let λ be a nonzero eigenvalue of $A^\top A$ with eigenvector $\mathbf{v}$. Then

$$\begin{aligned} A^\top A\mathbf{v} &= \lambda \mathbf{v} \\ \mathbf{v}^\top A^\top A\mathbf{v} &= \lambda \mathbf{v}^\top \mathbf{v} \\ \langle A\mathbf{v}, A\mathbf{v}\rangle &= \lambda \langle \mathbf{v}, \mathbf{v}\rangle \\ \left\|A\mathbf{v}\right\|^2 &= \lambda \|\mathbf{v}\|^2. \end{aligned}$$

We know that λ is nonzero, and $\mathbf{v}$ is nonzero so $\|\mathbf{v}\| > 0$. Hence $\left\|A\mathbf{v}\right\|$ is nonzero and thus positive. Thus

$$\lambda = \frac{\left\|A\mathbf{v}\right\|^2}{\|\mathbf{v}\|^2} \tag{14}$$

is the quotient of positive numbers and thus positive.

Now let $B := A^\top$ and note that $AA^\top = (A^\top)^\top (A^\top) = B^\top B$. Thus applying the same calculation to the rank-r matrix $B = A^\top$ obtains that $AA^\top$ is symmetric, that $\text{rank}(AA^\top) = r$, and that $AA^\top$ has exactly r nonzero eigenvalues, which are real and positive. □

A.2 Proof of Correctness of Construction of the Outer Product SVD

Proof of Correctness of Construction of Outer Product SVD, 8.7. We first claim that $\sigma_1 \geq \cdots \geq \sigma_r > 0$ are real. Indeed, by Section 8.1.3, the top r eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0$ of $A^\top A$ are real and positive, so their square roots $\sigma_1 \geq \cdots \geq \sigma_r > 0$ are real and positive.

Now we claim that$\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is an orthonormal set. This is immediately true by the Spectral Theorem, where the eigenvectors of $A^\top A$ produce an ordered set of $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$, so we get that $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is an orthonormal set.

Now we claim that $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ is an orthonormal set. Let $1 \leq i < j \leq r$. Then

$$\begin{aligned}
\langle \mathbf{u}_i, \mathbf{u}_j \rangle &= \langle \frac{1}{\sigma_i} A\mathbf{v}_i, \frac{1}{\sigma_j} A\mathbf{v}_j \rangle \\
&= \frac{1}{\sigma_i \sigma_j} \mathbf{v}_i^\top A^\top A \mathbf{v}_j \\
&= \frac{1}{\sigma_i \sigma_j} \sigma_i^2 \mathbf{v}_i^\top \mathbf{v}_j \\
&= 0
\end{aligned}$$

by orthonormality of the $\mathbf{v}_i$.

Now let $1 \leq i \leq r$. Then

$$\begin{aligned}
\|\mathbf{u}_i\|^2 &= \left\|A\mathbf{v}_i\right\|^2 \\
&= \frac{1}{\sigma_i^2} \left\|A\mathbf{v}_i\right\|^2 \\
&= \frac{1}{\sigma_i^2} \langle A\mathbf{v}_i, A\mathbf{v}_i \rangle \\
&= \frac{1}{\sigma_i^2} \sigma_i^2 \mathbf{v}_i^\top \mathbf{v}_i \\
&= \|\mathbf{v}_i\|^2 \\
&= 1
\end{aligned}$$

Thus $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ is an orthonormal set.

Finally, we claim that $A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$. This equality holds if and only if, for every $\mathbf{x} \in \mathbb{R}^n$, we have $A\mathbf{x} = \left(\sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top\right)\mathbf{x}$. We show the latter condition.

Since $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an orthonormal basis of $\mathbb{R}^n$, we may let $\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{v}_i$ for some constants $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. The eigenvectors corresponding to the 0 eigenvalues of $A^\top A$ — that is, $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$ — are an orthonormal basis for $\text{Null}(\mathbf{A}^\top \mathbf{A})$. We know that $\text{Null}(\mathbf{A}^\top \mathbf{A}) = \text{Null}(\mathbf{A})$, so $\{\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\}$ is an orthonormal basis for $\text{Null}(\mathbf{A})$. Thus

$A\mathbf{v}_{r+1} = \cdots = A\mathbf{v}_n = \mathbf{0}_m$. With this, we can compute

$$\begin{aligned}
\left(\sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top\right)\mathbf{x} &= \left(\sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top\right)\left(\sum_{i=1}^{n} \alpha_i \mathbf{v}_i\right) \\
&= \sum_{i=1}^{r}\sum_{j=1}^{n} \sigma_i \alpha_j \mathbf{u}_i \mathbf{v}_i^\top \mathbf{v}_j \\
&= \sum_{i=1}^{r}\sum_{j=1}^{n} \sigma_i \alpha_j \langle \mathbf{v}_j, \mathbf{v}_i \rangle \mathbf{u}_i \\
&= \sum_{i=1}^{r} \alpha_i \sigma_i \mathbf{u}_i \\
&= \sum_{i=1}^{r} \alpha_i A\mathbf{v}_i \\
&= \sum_{i=1}^{n} \alpha_i A\mathbf{v}_i \\
&= A\left(\sum_{i=1}^{n} \alpha_i \mathbf{v}_i\right) \\
&= A\mathbf{x}.
\end{aligned}$$

Thus $A = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ as desired. □

A.3 Proof of the Linear Algebra of the Full SVD

Proof of the Linear Algebra of the Full SVD, Theorem 8.2.

(i) Follows from the equivalence of outer product and full SVDs: $A = U\Sigma V^\top = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$.

(ii) Follows from the equivalence of outer product and full SVDs: $A = U\Sigma V^\top = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$.

(iii) Since $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ is the orthonormal set of columns of U, we have that $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ and $\{\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m\}$ are orthonormal bases for orthogonal subspaces. Since $\text{Span}(\mathbf{u}_1, \ldots, \mathbf{u}_r) = \text{Col}(\mathbf{A})$, we are left to show that $\text{Null}(\mathbf{A}^\top)$ is exactly the set of vectors orthogonal to $\text{Col}(\mathbf{A})$. However we know from note 3 that this is true.

(iv) Follows from applying (iii) to the SVD of $A^\top$, i.e., $A^\top = V\Sigma U^\top$.

(v) We have

$$\begin{aligned}
AA^\top &= (U\Sigma V^\top)(U\Sigma V^\top)^\top \\
&= U\Sigma V^\top V \Sigma U^\top \\
&= U\Sigma\Sigma U^\top.
\end{aligned}$$

(vi) We have

$$\begin{aligned} A^\top A &= (U\Sigma V^\top)^\top (U\Sigma V^\top) \\ &= V\Sigma^\top U^\top U\Sigma V^\top \\ &= V\Sigma^\top \Sigma V^\top. \end{aligned}$$

(vii) Follows from the equivalence of compact and full SVDs: $A = U\Sigma V^\top = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$.

□

A.4 Proof of Correctness of Construction of Full SVD

Proof of Correctness of Construction of Full SVD, 8.3. The only thing that has not already been shown from the proof of correctness of the outer product SVD algorithm is that U and V are square orthonormal matrices. But this is straightforward from Gram-Schmidt. □

A.5 Alternate Proof of the SVD

Our goal is to write

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top,$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthonormal matrices and Σ is a diagonal matrix. We want to answer two questions: under what conditions does this factorization exist, and if it does, how can we find it in closed form? We first handle the case where $\mathbf{A}$ is square and invertible, and then extend to the general case.

Claim. Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be invertible, and let $\mathbf{A}^\top\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^\top$ be an orthonormal diagonalization of $\mathbf{A}^\top\mathbf{A}$. Then $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$ is an SVD of $\mathbf{A}$, where

$$\mathbf{V} := \mathbf{P}, \qquad \Sigma := \sqrt{\mathbf{D}}, \qquad \mathbf{U} := \mathbf{A}\mathbf{V}\Sigma^{-1}. \tag{15}$$

Step 1: Diagonalize $\mathbf{A}^\top\mathbf{A}$. The matrix $\mathbf{A}^\top\mathbf{A}$ is symmetric, since

$$(\mathbf{A}^\top\mathbf{A})^\top = \mathbf{A}^\top(\mathbf{A}^\top)^\top = \mathbf{A}^\top\mathbf{A}.$$

So by the Spectral Theorem, an orthonormal diagonalization

$$\mathbf{A}^\top\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^\top$$

exists, where the columns of $\mathbf{P}$ are orthonormal eigenvectors of $\mathbf{A}^\top\mathbf{A}$ and $\mathbf{D}$ is a diagonal matrix of the corresponding eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, ordered from largest to smallest. Since $\mathbf{A}$ is invertible, $\operatorname{rank}(\mathbf{A}^\top\mathbf{A}) = n$, so all n eigenvalues are positive (see the proof of the eigenvalues of $\mathbf{A}^\top\mathbf{A}$ above). Thus

$$\sqrt{\mathbf{D}} = \begin{bmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_n} \end{bmatrix}$$

is a real, invertible diagonal matrix.

Step 2: Construct U, Σ, and V. Take $\mathbf{V} = \mathbf{P}$ and $\Sigma = \sqrt{\mathbf{D}}$. The formula for $\mathbf{U}$ comes from rearranging the factorization we want. Multiplying on the right by $\mathbf{V}$ (using $\mathbf{V}^\top\mathbf{V} = \mathbf{I}$) and then by Σ^{-1} gives

$$\begin{aligned}\mathbf{A} &= \mathbf{U}\Sigma\mathbf{V}^\top \\ \mathbf{AV} &= \mathbf{U}\Sigma\mathbf{V}^\top\mathbf{V} = \mathbf{U}\Sigma \\ \mathbf{AV}\Sigma^{-1} &= \mathbf{U}.\end{aligned}$$

Step 3: Check that this is an SVD. $\mathbf{V} = \mathbf{P}$ is orthonormal and Σ is diagonal by construction. To see that $\mathbf{U}$ is orthonormal, note that Σ^{-1} is diagonal (so $(\Sigma^{-1})^\top = \Sigma^{-1}$), that $\mathbf{D} = \Sigma^2$, and that

$$\mathbf{V}^\top\mathbf{A}^\top\mathbf{AV} = \mathbf{P}^\top\mathbf{PDP}^\top\mathbf{P} = \mathbf{D}.$$

Then

$$\begin{aligned}\mathbf{U}^\top\mathbf{U} &= (\mathbf{AV}\Sigma^{-1})^\top(\mathbf{AV}\Sigma^{-1}) \\ &= \Sigma^{-1}\mathbf{V}^\top\mathbf{A}^\top\mathbf{AV}\Sigma^{-1} \\ &= \Sigma^{-1}\mathbf{D}\Sigma^{-1} \\ &= \Sigma^{-1}\Sigma^2\Sigma^{-1} \\ &= \mathbf{I}.\end{aligned}$$

Finally, the factorization holds, since $\mathbf{V}$ is square and orthonormal so $\mathbf{VV}^\top = \mathbf{I}$:

$$\begin{aligned}\mathbf{U}\Sigma\mathbf{V}^\top &= \mathbf{AV}\Sigma^{-1}\Sigma\mathbf{V}^\top \\ &= \mathbf{AVV}^\top \\ &= \mathbf{A}.\end{aligned}$$

Since the diagonalization $\mathbf{A}^\top\mathbf{A} = \mathbf{PDP}^\top$ always exists, we can always find $\mathbf{U}$, Σ, and $\mathbf{V}$ through the construction above: $\mathbf{V}$ is the matrix of orthonormal eigenvectors of $\mathbf{A}^\top\mathbf{A}$, Σ is the square root of the diagonal matrix in the diagonalization of $\mathbf{A}^\top\mathbf{A}$, and $\mathbf{U} = \mathbf{AV}\Sigma^{-1}$.

The general case. In general, $\mathbf{A} \in \mathbb{R}^{m\times n}$ has rank $r \le \min\{m,n\}$, so the eigenvalues of $\mathbf{A}^\top\mathbf{A}$ satisfy $\lambda_1 \ge \cdots \ge \lambda_r > 0$ and $\lambda_{r+1} = \cdots = \lambda_n = 0$. If $r < n$, then $\sqrt{\mathbf{D}}$ is not invertible, and $\mathbf{U} = \mathbf{AV}\Sigma^{-1}$ is undefined. The fix is to keep only the r positive eigenvalues. Split

$$\mathbf{V} = \begin{bmatrix}\mathbf{V}_r & \mathbf{V}_{n-r}\end{bmatrix} \qquad \text{and} \qquad \Sigma_r = \begin{bmatrix}\sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_r}\end{bmatrix},$$

where the columns of $\mathbf{V}_r$ are the eigenvectors for the r positive eigenvalues, and let

$$\mathbf{U}_r := \mathbf{AV}_r\Sigma_r^{-1} \in \mathbb{R}^{m\times r}.$$

The same computation as in Step 3 shows $\mathbf{U}_r^\top \mathbf{U}_r = \mathbf{I}_r$. Each column $\mathbf{v}$ of $\mathbf{V}_{n-r}$ is an eigenvector of $\mathbf{A}^\top\mathbf{A}$ with eigenvalue 0, so

$$\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A}\mathbf{v} = 0,$$

i.e., $\mathbf{A}\mathbf{V}_{n-r} = \mathbf{0}$. Therefore

$$\begin{aligned}
\mathbf{A} &= \mathbf{A}\mathbf{V}\mathbf{V}^\top \\
&= \mathbf{A}\mathbf{V}_r\mathbf{V}_r^\top + \mathbf{A}\mathbf{V}_{n-r}\mathbf{V}_{n-r}^\top \\
&= \mathbf{A}\mathbf{V}_r\mathbf{V}_r^\top \\
&= \mathbf{U}_r\Sigma_r\mathbf{V}_r^\top, \qquad (16)
\end{aligned}$$

which is the compact SVD of $\mathbf{A}$. To get the full SVD, extend $\mathbf{U}_r$ to a square orthonormal matrix $\mathbf{U}$ using Gram-Schmidt, and pad Σ_r with zeros.